

Skills

Technical Leadership, AI/ML Platforms Design, High Load Systems, Recommenders.

Languages C/C++, Python, Java, Go, SQL.

Experience

2024 Apr – **Senior Staff Software Engineer / TLM, Google - Search AI Mode Growth.**

2026 Jan Founded and led Search AI Notifications from pre-Gemini infrastructure to agentic serving stack, growing engagement from -6% to +22%, accounting for 57% of inorganic Search AI Mode growth.

- o Built and grew an 18-person team owning a large-scale production platform; navigated sister team layoffs while beating growth targets 3x.
- o Landed real-time (15-minute fresh) transformer recommendation model training and experimentation (+32% engagement).
- o Delivered significant infrastructure optimizations in 2024 to self-fund growth.
- o Built GenAI infrastructure in collaboration with Gemini and launched:
 - Topic updates generation: users receive meaningful updates on followed queries (+16% re-engagement).
 - [Samsung Now Bar](#): real-time sports commentary generated from UGC, delivered as Android Live Activity.
 - [Daily Listen](#): daily AI-generated podcasts personalized to user interests.
- o Redefined serving stack as a collection of agentic tool calls with sub-1-day time to production.
- o Grew team leads: landed 3 promotions to Staff Engineer in 2 years.

2021 Apr – **Staff Software Engineer, Google - Search.**

2024 Apr Led a unified Search Growth Infrastructure team delivering >20% of inorganic Search growth in 2023. Established cross-org collaboration across 87 engineers from Google Research, Core Infrastructure, and Android — modernizing user-content matching with transformer-based retrieval and LLM-powered generation.

- o Reduced new product idea-to-MVP time from a month to 2 hours and time to launch from a quarter to 2 weeks.
- o Built a full-cycle ML/AI ecosystem (indexing, generation, matching, training, serving, calibration):
 - Augmented heuristic content-user matching with ML-based retrieval, improving user engagement by 64%.
 - Created scalable architecture supporting item2item, ScaNN, and custom interest retrieval strategies (+10% engagement in first quarter).
- o Reduced model staleness from 1 month to 1 day via real-time training with lifetime user history.
- o Introduced on-device ML, improving notification triggering quality by 7%.

2018 Apr – **Senior Software Engineer, Google - Search.**

2021 Apr Envisioned and led the team implementing ML training and serving ecosystem for Search's News and Content Discovery.

- o Designed and led the implementation of unified ML infrastructure (C++) for Discover, Notifications, Suggest, and user modeling teams featuring:
 - zero training/serving skew near-real-time training data generation system, reducing feature engineering work by 70%.
 - dynamic triggering rate optimization based on recent and long-term user activity patterns,
 - reinforcement learning support (in collaboration with DeepMind).
- o Scaled inference serving infrastructure to handle 24x more inferences on the same resources.
- o Launched flexible GPU/TPU resource acquisition and scheduling solution, improving accelerator utilization by 140% (in collaboration with DeepMind).
- o Restructured Discover Search training to support TPU, achieving 40x speedup (Python).
- o Introduced reliability culture to ML ranking, establishing oncall rotation and making it a mission-critical product component.

2016 Apr – **Software Engineer, Google - YouTube.**

2018 Apr Built storage abstraction microservices framework with no-downtime migration support (C++) and improved YT Public Data API performance, usability, and security (Python).

- o Designed and implemented storage management framework allowing the user to create a full set of standard RESTful APIs for a defined resource in several hours regardless of the storage type (C++).
- o Migrated video metadata migration between Vitess and Spanner with no downtime, data loss, or client latency increase (C++).

Sep 2015 – **Fellow, Insight Data Science.**

Oct 2015 Built [CityGravity.info](#), a trip schedule optimization app to see the greatest number of attractions.

2010 Aug – **Graduate Researcher, University of Arizona, Tucson, AZ.**

2016 Jan Developed Fortran code modeling the molecular system total wave function as an optimal combination of explicitly correlated Gaussian functions. Generalized the approach to be applied to exotic atoms and molecules and extended it to excited states. The results were published in over a dozen peer-reviewed papers and presented on multiple physics and supercomputing conferences.

Jan 2010 – **Intern, Joint Institute of Nuclear Research, Dubna, Russia.**

May 2010 Optimized muon detector signal processing code for the CMS experiment by vectorizing the core algorithm.

Education

2010–2015 **Ph.D.**, *University of Arizona*, Tucson, AZ – Physics.

2005–2010 **B.S. (Hons.)**, *Saratov State University*, Saratov, Russia – Physics.